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in

COMPUTER SCIENCE AND ENGINEERING

iTantra

*Indian Multilingual TTS & STT Aided Neural Transceiver
Radio Access for Low Bitrate Links*

Smart India Hackathon 2026 · Problem Statement **SIH26173**

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ABSTRACT

In alert and distress situations, transmitting voice is more inclusive than transmitting written text, because it does not require either party to be literate. Voice, however, is data-intensive. Intelligible speech requires about 256 kbps as raw audio, and even Opus, the most efficient open-source speech codec available, has a practical intelligibility floor near 6 kbps. The narrowband radio links that keep working when terrestrial infrastructure has failed — LoRa, HF data links and narrowband satellite bursts — deliver only a few hundred bits per second. Voice over such links is therefore not merely difficult but bounded away from possibility, because audio codecs compress the waveform and an intelligible waveform has an irreducible size.

iTantra resolves this by removing the waveform from the link. Speech is recognised on the sending handset, the resulting text is transmitted as a compact binary frame of a few dozen bytes, and speech is re-synthesised on the receiving handset. Both operators only speak and listen; neither reads nor types. Ten Indian languages — Hindi, Gujarati, Marathi, Kannada, Malayalam, Tamil, Telugu, Odia, Bengali and English — run entirely on the device, with no cloud service, no SIM, no proprietary SDK and no internet permission in the application manifest.

Compression proceeds in three graceful stages: semantic transcoding of speech to text, script packing that represents each Indic character in one byte instead of the three UTF-8 requires, and template coding that maps common operational sentences to single-byte identifiers. A three-second Hindi sentence occupying 96 000 bytes as raw audio is transmitted in 44 bytes, a reduction of 2 182 to 1. Because every device holds the template table in all ten languages, a template-coded alert is announced in whichever language the receiver has selected, giving cross-language operation with no translation model.

Keywords: speech recognition, speech synthesis, semantic compression, low-bitrate communication, Indian languages, on-device inference, LoRa, disaster communication.

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1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background

Communication is the first casualty of a disaster. When mobile towers fail, the equipment that keeps working is the handheld radio set, because it needs no infrastructure at all — only a battery and an antenna. Such a radio provides a narrowband data channel of a few hundred bits per second, fixed by terrain, power budget and spectrum allocation. A voice call cannot cross it.

This is not a limitation a better codec will remove. Reproducing a human voice with enough fidelity for words to stay distinguishable needs 16 000 samples per second at 16 bits, which is 256 000 bits for every second of speech. Opus reduces this to about 6 000 bits per second, below which intelligibility collapses. The remaining gap to a narrowband link is a factor of twenty, and it is a bound rather than an engineering deficiency.

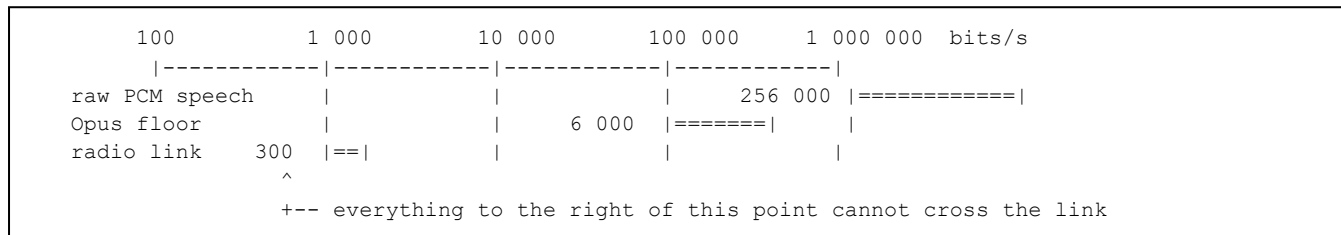


Figure 1.1: Data rates on a logarithmic scale. Both audio representations lie far beyond the capacity of the link.

1.2 Problem Statement

Problem Statement SIH26173 of the Indian Space Research Organisation requires an Android application with lightweight, accurate speech-to-text and text-to-speech models for ten Indian languages, running locally on a low-power device. The speech-to-text module must activate after detecting pauses and stoppages, form the sentences detected, and stream the data efficiently through a Wi-Fi or Bluetooth connected embedded device or another handset running the same application. The text-to-speech module must convert received text into intelligible speech played as a voice note, and alert messages must be announced at highest volume, non-interruptible. Two handsets running the same application must work as a walkie-talkie using push-to-talk, and as a telephone when that is turned off.

1.3 Objectives

1. Convert speech to text on the sending device and text back to speech on the receiving device, so both users interact only by speaking and listening.
2. Support ten Indian languages with quantised on-device models under 70 MB per pack.
3. Define a binary wire protocol carrying a full sentence in under 60 bytes.
4. Trigger recognition from natural pauses while holding idle CPU below two per cent.
5. Provide a transport abstraction including a serial-shaped path for a radio module.
6. Deliver alerts at maximum volume on a locked and silenced handset.
7. Instrument the application so accuracy, latency and efficiency are measured, not claimed.

1.4 Scope and Limitations

Every device runs both halves of the system; the role is a runtime mode, not a build. Operation is entirely offline. Speaker identity, prosody and background audio are discarded with the waveform, and are compensated by a sender identifier, an explicit alert priority class and a distinct synthesis voice per node. Cross-language delivery applies to template-coded messages only. Code-mixed Hindi–English speech is a known weakness, disclosed together with its mitigation.

2. LITERATURE SURVEY AND EXISTING SYSTEMS

2.1 Speech Recognition for Indian Languages

Speech recognition has moved from hidden Markov models to end-to-end neural architectures. The Conformer architecture of Gulati et al. combines convolutional blocks, which capture local acoustic events such as plosive bursts and formant transitions, with self-attention blocks supplying sentence-level context. It has become the standard streaming encoder because it reaches competitive accuracy at a cost acceptable on mobile hardware.

For Indian languages the AI4Bharat initiative at IIT Madras has released the IndicConformer acoustic models together with the IndicSUPERB, Kathbath and IndicVoices corpora. These matter here because multilingual models trained mainly on European languages perform poorly on Indic phonology and on scripts such as Odia and Kannada. Vosk offers smaller streaming models with immediate Android support and lower accuracy, useful as a development baseline. Whisper is accurate but is not a streaming architecture and is heavy for entry-tier processors.

2.2 On-Device Speech Synthesis

Earlier synthesis pipelines needed two models, text to spectrogram and spectrogram to waveform, and were too slow for phones. The VITS architecture of Kim, Kong and Son performs both in a single pass and includes a duration predictor, the component responsible for natural rhythm. The Piper project provides VITS voices already exported to ONNX, and AI4Bharat provides Indic-TTS voices of higher naturalness. Notably, two of the four stages of a synthesis pipeline are conventional software engineering rather than machine learning: text normalisation, which decides whether digits are a quantity or an identifier, and phonemisation, which is far more reliable for the largely phonetic Indic scripts than for English.

2.3 Messaging over Low-Bitrate Radio

Several open-source systems already carry text over long-range radio. Meshtastic implements text messaging over LoRa with an Android client connected by Bluetooth, and informs the design of framing, addressing, relay and duplicate suppression. Briar demonstrates robust Bluetooth and Wi-Fi transport on Android, including discovery, reconnection and degraded states. APRSdroid connects Android to amateur radio over a Bluetooth serial profile, and is a working instance of the deployment path adopted here.

2.4 The Gap

The common property of these systems is that the user types the message and reads the reply. They therefore solve the transport problem and leave the accessibility problem untouched, which is exactly the gap the problem statement identifies.

System	Strength	Limitation for this problem
Opus voice codec	Preserves the speaker's voice; no model needed	Cannot go below ~6 kbps; unusable on a 300 bps link
Meshtastic / LoRa text	Works comfortably within the bandwidth	Requires typing and reading; excludes non-literate users
Cloud speech APIs	High accuracy, many languages	Require internet; prohibited by the problem statement
Satellite voice terminals	Global coverage	Cost and licensing place them beyond field teams
iTantra (proposed)	Voice at both ends over a 300 bps link	Speaker identity and prosody are not preserved

Table 2.1: Comparison of existing approaches with the proposed system.

The contribution of this project is therefore not a new speech model or a new radio protocol, but the composition of the two together with a compression scheme that exploits a property unavailable to general-purpose systems: because both endpoints are controlled and the frame header declares the language, Indic text packs into a single-byte alphabet and common sentences reduce to single-byte identifiers.

3. PROPOSED SYSTEM

3.1 Principle

The proposed system removes the waveform from the link. Speech is recognised on the sending handset and converted to text; the text is compressed and framed; the frame crosses the radio link; and the text is re-synthesised as speech on the receiving handset. The interface offered to both operators is voice, and the representation carried by the link is text. Everything discarded — the speaker's timbre, the wind behind him, the exact pitch contour, the room acoustics — is not the message but the carrier the message arrived in.

3.2 Three Levels of Compression

The system does not use one compression scheme. It degrades gracefully through three, choosing the most aggressive that the situation allows.

Level 1 — Semantic transcoding. Speech becomes UTF-8 text. This alone gives a reduction of roughly 770 to 1, and everything below it is refinement.

Level 2 — Script packing. UTF-8 encodes Devanagari, Tamil, Kannada and the other Indic scripts at three bytes per character, because it must represent every script at once. The frame header already declares the language, so the receiver knows which 128-codepoint Unicode block applies; subtracting the block base gives one byte per character. Characters outside that block — Latin digits, punctuation, and the zero-width joiners that control conjunct formation — are carried by an escape sequence, so the transformation stays lossless.

Level 3 — Template codes. In an emergency the vocabulary is small and predictable. A shared table maps common operational sentences to one-byte identifiers. After recognition the text is fuzzy-matched against this table, and on a confident match the identifier alone is sent. Because every device holds the table in all ten languages, the receiver renders the sentence in whichever language it has selected — cross-language operation with no translation model, no extra download and no extra latency.

Representation	Size	On a 300 bps link	Note
Raw PCM, 16 kHz, 16-bit mono	96 000 B	43 min	What the microphone produces
Opus at 16 kbps	6 000 B	2.7 min	Good quality voice call
Opus at 6 kbps	2 250 B	60 s	Thin, robotic, the practical floor
UTF-8 text	99 B	2.6 s	What the recogniser produces
Packed text with header (Level 2)	44 B	1.2 s	One byte per Indic codepoint
Template code with header (Level 3)	13 B	0.3 s	One byte of payload

Table 3.1: *A three-second Hindi sentence in each representation.*

3.3 Advantages

1. **Inclusive by construction.** Neither operator reads or types, so the system works for a person who cannot read, for an operator wearing gloves, and with the screen off.
2. **Feasible where voice is not.** A sentence crosses a 300 bps link in 1.2 seconds; the same sentence as Opus audio would need 60 seconds.
3. **Entirely offline.** The application declares no internet permission, so the property is enforced by the operating system rather than asserted by the developers.
4. **Deployable on existing hardware.** Long-range operation adds about ₹1 500 of radio hardware per pair and no change to the handset software.

3.4 Feasibility

Every component exists today, is open source and runs on ARM; nothing requires research. There is no server, no data plan and no per-message charge. The operating model is that of a walkie-talkie, already familiar to the intended users.

4. SYSTEM ARCHITECTURE AND DESIGN

Every device runs the complete system. There is no dedicated transmitter and no dedicated receiver; the role is a runtime mode, and a single application package is installed on both handsets.

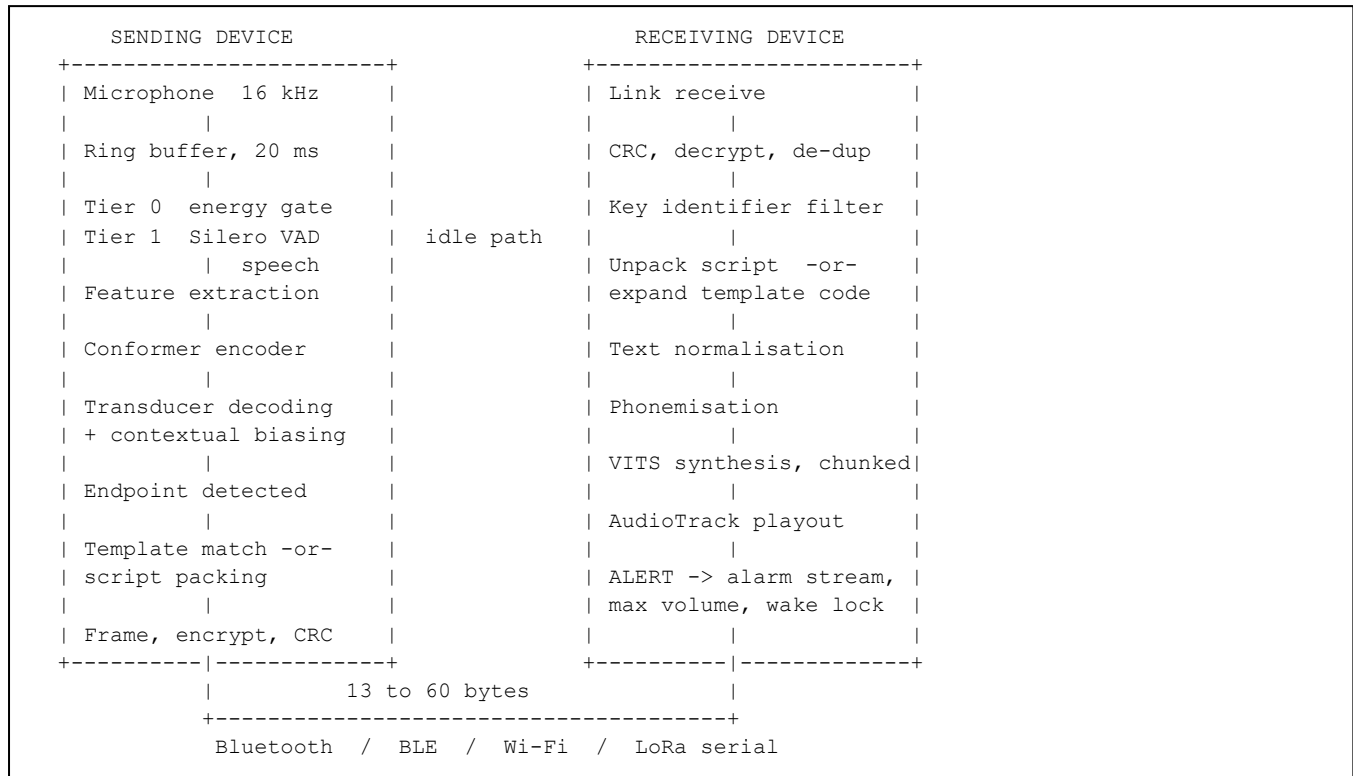


Figure 4.1: End-to-end signal path. Both halves exist on every device.

4.1 Modules

Module	Responsibility
app	Interface, push-to-talk, alert delivery, pairing, instrumentation, foreground service
core-audio	Capture and playback, ring buffers, alert audio policy, audio focus
core-asr	Three-tier detection, endpointing, recognition, biasing, confidence
core-tts	Normalisation, phonemisation, chunked synthesis, streaming playout
core-link	One transport interface with four implementations
core-proto	Frame codec, CRC, encryption, script packing, template tables, replay window
core-models	Language pack manifest, checksum verification, storage accounting
bench	Word error rate harness, latency logging, noise mixing, scorecard export

Table 4.1: Module responsibilities.

4.2 Threading

Audio work never touches the interface thread and the recogniser never blocks capture. Four threads run with bounded queues between them. The capture thread runs at urgent audio priority and performs no memory allocation, since a dropped block is a lost utterance. The inference thread drains the recognition queue. The link thread performs blocking input and output. The synthesis thread generates chunks ahead of the playback clock. The engine lives in a foreground service, because the normal operating state of a radio is screen-off.

The most important boundary in the design is that **core-proto carries no Android dependency**. The highest-risk component of the system, the frame codec, is therefore unit tested and fuzz tested on a desktop in seconds rather than on a handset in minutes, and the whole transport layer can be proven before any model exists.

5. WIRE PROTOCOL

Ten bytes of header and two of trailer buy priority, relay, integrity and authentication. The payload is what remains. All multi-byte fields are most significant byte first.

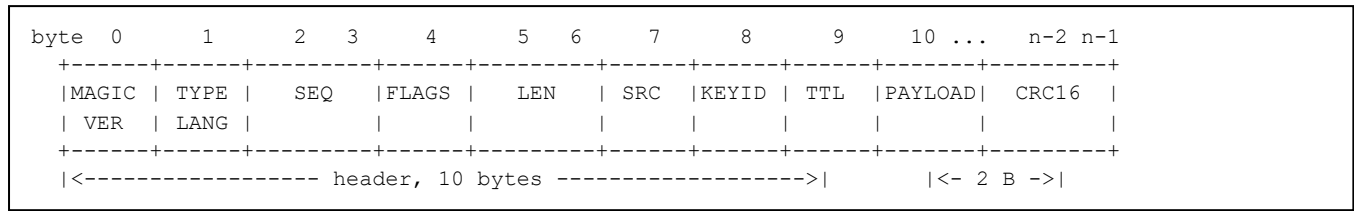


Figure 5.1: *Frame format.*

Field	Bytes	Description
MAGIC / VER	1	Sentinel for resynchronising after corruption, plus protocol version
TYPE / LANG	1	Message type, and language index 0–9
SEQ	2	Per-sender sequence number driving acknowledgement and the replay window
FLAGS	1	Final, partial, encrypted, fragment, packed, template, two confidence bits
LEN	2	Payload length; the basis of framing over a byte stream
SRC	1	Sender node identifier
KEYID	1	First byte of the hash of the shared key; a fast rejection filter
TTL	1	Remaining relay hops
PAYLOAD	n	Packed text, template identifier, coordinates or control data
CRC16	2	CCITT checksum over header and payload

Table 5.1: *Frame fields.*

5.1 Addressing

Radio does not route; it broadcasts. There is no destination field: every frame reaches every device holding the shared key, exactly as a walkie-talkie does. Selection is by cryptographic key rather than by address, because address filtering is a convention that any transmitter may disregard.

5.2 Integrity and Authentication

The problem statement does not ask for this. It is included because the system can wake a locked, silenced handset and announce at maximum volume on command; if it accepted unauthenticated input, any transmitter in radio range could inject a fraudulent evacuation order into a disaster area. Frames are protected by AES-256 in Galois/Counter Mode with a pre-shared key held by every paired unit, with the whole header as associated data so the sender identifier cannot be altered. A sliding window of sixty-four sequence numbers per sender prevents a recorded alert being replayed. The key is transferred optically by QR code and never crosses the radio medium.

5.3 Reliability

Corrupt frames are discarded and never spoken, since a garbled instruction is more dangerous than a missing one. Alerts are acknowledged and retried up to three times at 300 ms; ordinary conversation is fire-and-forget, because retransmitting stale speech is worse than losing it. A heartbeat every two seconds distinguishes a quiet channel from a dead one, and undelivered frames queue locally and flush on reconnection.

6. IMPLEMENTATION

6.1 Recognition

Running the acoustic model continuously would exhaust the battery in about two hours and fail the efficiency criterion outright. Detection is therefore tiered, so the expensive stage runs only when the cheap stages agree that a human is speaking.

Tier	Mechanism	Cost / 20 ms	Purpose
0	Energy gate against an adaptive noise floor	~3 μ s	Rejects silence, which is most of elapsed time
1	Silero VAD, 1.8 MB	~300 μ s	Separates speech from wind, engines and door slams
2	Conformer acoustic model	~4 ms	Recognition; runs only when tier 1 agrees

Table 6.1: *Three-tier voice activity detection.*

Endpointing is a state machine. In telephone mode an utterance finalises after 400 ms of trailing silence; in push-to-talk mode the key release is itself an explicit end-of-utterance signal and the window falls to 150 ms, which is why push-to-talk is measurably faster. A 250 ms leading pad is kept from the ring buffer so the first phoneme is never clipped.

Accuracy on critical vocabulary is raised by contextual biasing, which boosts decoding scores for a supplied phrase list — distress terms, call signs, sector numbers and place names. Negation terms are weighted heavily, because an error turning “do not evacuate” into “now evacuate” inverts the meaning and is the most dangerous failure the system can produce.

Precision	Model size	Relative speed	Effect on WER	Decision
float32, as trained	120 MB	1.0×	baseline	Not shippable
float16	60 MB	1.3×	negligible	Intermediate
int8 dynamic	32 MB	2.5×	+1 % relative	Adopted

Table 6.2: *Quantisation is what makes on-device speech viable on an entry-tier handset.*

6.2 Synthesis

Text normalisation runs first and is a rule file rather than a model: it decides that 112 is dialled digit-wise while “5 km” is a quantity with an expanded unit. This distinction is the difference between a synthesiser that sounds professional and one that sounds broken. VITS then produces audio directly from phonemes in a single pass at a real-time factor near 0.15. Rather than synthesising the whole sentence and then playing it, the sentence is split at clause boundaries and playback starts on the first chunk, cutting time-to-first-audio from about 700 ms to about 180 ms for the same total work.

6.3 Transport

Property	Bluetooth	Bluetooth LE	Wi-Fi	LoRa serial
Range	10–30 m	10–50 m	50–150 m	2–15 km
Usable rate	200 kbps	5–20 kbps	>10 Mbps	0.3–5 kbps
Carries a voice call	Marginally	No	Yes	No
Carries an iTantra frame	Yes	Yes	Yes	Yes
Role	Default	Standby	Reach	Deployment

Table 6.3: *The four transports. Read the last two rows together: on the link that works when infrastructure fails, audio is impossible and iTantra is comfortable.*

Bluetooth RFCOMM is the default because it is a reliable ordered byte stream and the same profile an ESP32 or radio modem presents — the code that talks to another phone is the code that talks to a radio. Deployment changes only what sits at the far end of the link.

7. TESTING, EVALUATION AND RESULTS

The problem statement allocates eighty per cent of the assessment to three measurable properties: accuracy at 40 %, latency at 20 % and efficiency at 20 %. The application is therefore instrumented from the first commit and exports its own results, so the submission is a set of measurements rather than a set of claims.

Status of these figures. Implementation is in progress. The frame sizes below follow arithmetically from the format in Section 5 and are exact. The accuracy, latency and efficiency figures are stated as **design targets**, not measurements, and the conditions under which each will be measured are defined below.

7.1 Measurement Conditions

Figures are reportable only from the target handset, in a release build, after a thirty-minute thermal soak, over at least a hundred utterances, as median and 95th percentile rather than best case. Entry-tier handsets throttle after about ten minutes of inference and the real-time factor can double, so sustained figures are more defensible. Word error rate uses held-out public corpora — IndicSUPERB, Kathbath, IndicVoices and FLEURS — never self-recorded samples.

7.2 Results and Targets

Frame	Total	At 300 bps	Compression vs raw audio
Template alert	13 B	0.3 s	7 385 : 1
Packed sentence	44 B	1.2 s	2 182 : 1
Packed sentence, authenticated	60 B	1.6 s	1 600 : 1
Equivalent Opus audio	2 250 B	60 s	— (43 : 1 against iTantra)

Table 7.1: Frame sizes, derived exactly from the protocol. All four ratios are reported, since the comparison against a codec is the honest one and the comparison against raw audio is the striking one.

Accuracy target	WER	Latency / efficiency target	Value
Clean, read speech	< 12 %	End to end, push-to-talk	500–800 ms
+20 dB SNR	< 15 %	End to end, telephone mode	750–1100 ms
+10 dB SNR	< 22 %	Time to first synthesised audio	150–250 ms
+5 dB SNR	< 32 %	Installer, two languages	< 30 MB
Critical vocabulary, biased	< 6 %	Idle CPU while listening	< 2 %

Table 7.2: Accuracy, latency and efficiency targets. The critical-term figure matters most: overall word error rate is a poor proxy for usefulness, and what decides whether a message is actionable is whether the distress terms and sector numbers survive.

7.3 Test Cases

No.	Test	Expected result
1	Encode and decode 10 000 random frames	Every frame recovered exactly
2	Pack and unpack every codepoint of each Indic block	Original string recovered exactly
3	Feed the decoder truncated and bit-flipped input	No crash; recovery on the next valid frame
4	Alter one byte of an authenticated frame	Verification fails; frame discarded
5	Speak ten Hindi sentences in aeroplane mode	All delivered; median delay under 800 ms
6	Alert to a locked, silenced handset	Announced at full volume; prior volume restored
7	Template alert, Tamil sender to Hindi receiver	Correct Hindi sentence announced
8	Run continuously for thirty minutes	Real-time factor under twice the initial value

Table 7.3: Principal test cases.

8. CONCLUSION, FUTURE SCOPE AND REFERENCES

8.1 Conclusion

This project has presented the design and specification of iTantra, an Android application providing voice-to-voice communication over radio links whose capacity cannot carry a voice call. The central idea is that the waveform, which is the overwhelming majority of the data in a speech signal, is not the message but the carrier the message arrived in. By recognising speech on the sending device and re-synthesising it on the receiving device, the system transmits meaning rather than sound, and a three-second sentence occupying 96 000 bytes as audio is carried in 44 bytes.

The system is symmetric, entirely offline, and built only from open-source components, with two restrictively licensed dependencies — `espeak-ng` under GPL-3.0 and Meta MMS under CC-BY-NC — disclosed rather than left to be discovered. It supports ten Indian languages, targets low and mid-range handsets, and is instrumented so that accuracy, latency and efficiency are measured rather than asserted.

It is worth stating plainly that iTantra is not a speech-recognition demonstration with a Bluetooth link attached. It is a communications system that uses speech models as its codec. That framing matches the language of the problem statement itself — a neural transceiver providing radio access for low bitrate links — and it is the framing under which the design decisions in this report are coherent rather than arbitrary.

8.2 Future Scope

1. **Completion and measurement.** Implement the specified system and replace every design target in Section 7 with a measured result from the evaluation harness.
2. **Speaker-adaptive synthesis.** Transmitting a speaker embedding of a few bytes would let the receiver approximate the original voice without sending any audio.
3. **Free-form cross-language delivery.** Cross-language operation is presently limited to template messages; a compact on-device translation model would extend it.
4. **Mesh networking.** A routing protocol would improve delivery across many nodes.
5. **Satellite integration.** The frame size suits narrowband satellite burst channels, which would extend coverage beyond terrestrial radio range entirely.
6. **Further languages.** Adding a language needs no code change, only an additional pack of model, voice, rules and template table.

8.3 References

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